which they live. Later ecologists developed mathematical models to forecast population dynamics within ecosystems. Evolutionary theories also advanced with the discovery of the structure of DNA, and the evolutionary "vehicle" provided by mutation as DNA is replicated.

New frontiers

Improved technology opened up new possibilities for ecology. An electron microscope can now make images to half the width of a hydrogen atom, and computer programs can analyze the sounds made by bats and whales. Camera traps and infrared detectors film nocturnal creatures, and tiny satellite devices fitted to birds can track their movements.

Laboratory analysis of the DNA of feces, fur, or feathers indicates which species an animal belongs to, and throws light on the relationship between different organisms. It is now easier than ever for ecologists to collect data, helped by a growing army of citizen scientists.

New concerns

Early ecology was driven by a desire for knowledge. Later, it was used to find better ways to exploit the natural world for human needs. As time went on, the consequences of this exploitation have become increasingly evident. Deforestation was highlighted as a problem as early as the 18th century, and the problems of air and water pollution became obvious in industrialized nations in the 19th century. In 1962, Rachel Carson's book *Silent Spring* alerted the world to the dangers of pesticides, and six years later Gene Likens demonstrated the link between power station emissions, acid rain, and fish deaths.

In 1985, a team of Antarctic scientists discovered the dramatic depletion of atmospheric ozone over Antarctica. The link between greenhouse gases and a warming of Earth's lower atmosphere had been made as early as 1947